FIRMWARE UPLOAD GUIDELINE

1. TOOLS REQUIRED

a. J Link Debug Probe
b. EtherCAT Interface Board and Control Board
c. J link Command software (The link is for 64-bit Windows)

2. STEP BY STEP INSTRUCTIONS

1) Download and install J link documentation pack. Inside the pack, **J link commander** will also be installed on the system.
2) Once the software is installed, proceed to make the hardware connections between the **EtherCAT Boards** and the **J link Debug Probe** as described below

Table 1 Pinout connections between Jlink Probe and EtherCAT board

S.No	J Link Debug Probe Pin	EtherCAT Board Pin
1	VTref	VCC_3V3
2	SWDIO	SWDIO
3	SWCLK	SWDCLK
4	SWO (Optional)	SWO (Optional)
5	RESET	RST
6	GND	GND

3) Images are also shared from both the debug probe and the board for easiness

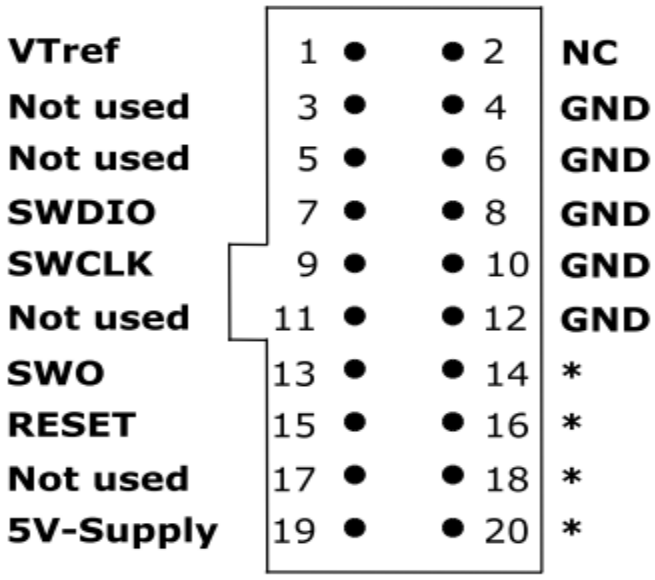


Figure 1 JLink Probe pinout information

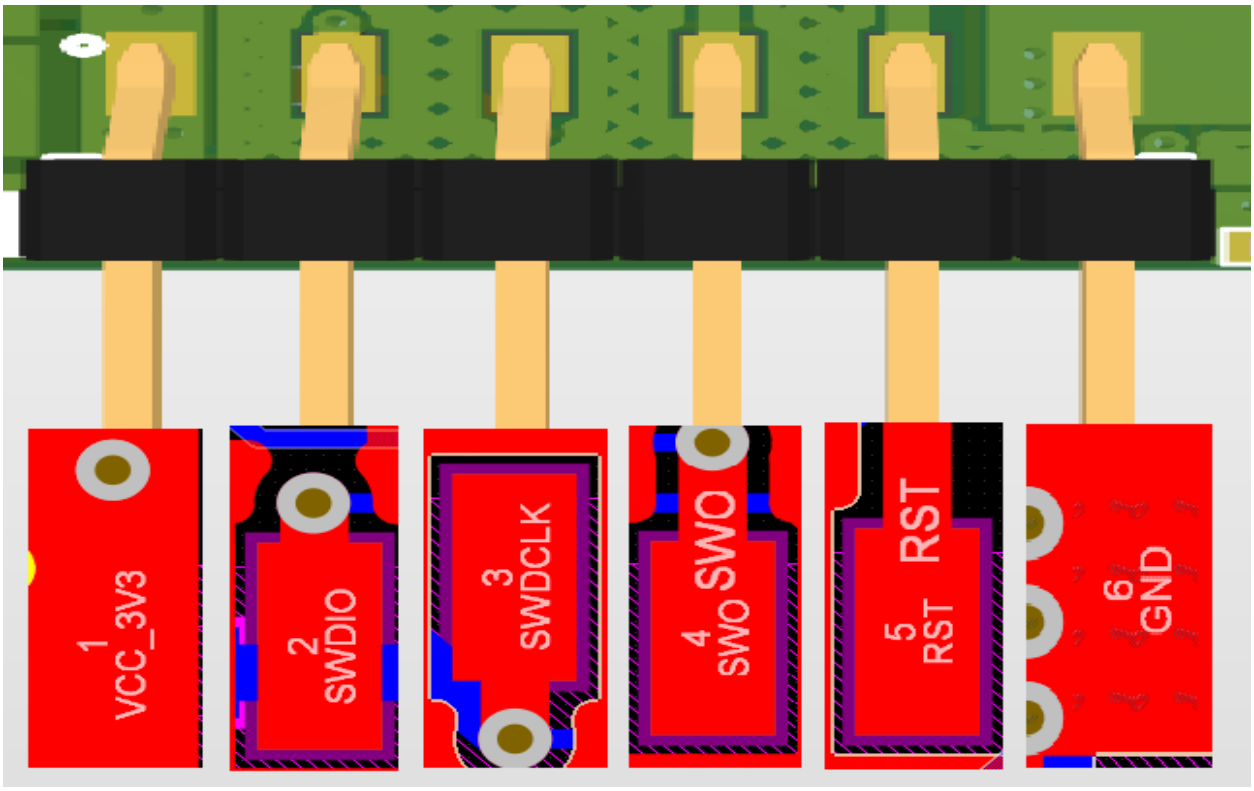


Figure 2 EtherCAT Board pinout information

4) Once the connections are established, power up the PCB board using the headers on **Interface Board** with an adequate power supply.
5) Connect the J link to the target PC and then afterwards open **Jlink commander** software.
6) Verify VTref are being read as **3.3V** to make sure the board is properly connected to the **Jlink Probe** as shown in **Figure 3**

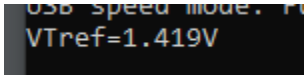


Figure 3 Screenshot showing VTref for unconnected Jlink

7) Type **connect** on the command line interface to establish connection with the custom EtherCAT board.
8) Once the connection is established, an interface may open where it will ask for specific MCU part number, proceed to select **ATSAME53J18** from the list and click **OK**.
9) Another prompt will ask for type of Interface? (SWD, JTAG etc). Type **S** and press **Enter** to proceed.
10) Choose default for the rest of the options by pressing **Enter**.
11) Once finished, type **loadfile <path\to\hex\file>**
12) Press **Enter** and the program will start to load. Once the program has loaded successfully, an **On-screen** message will indicate success.
13) Disconnect the **Jlink Probe** and power cycle the board (turn the power OFF / ON) and start testing!